Meshtastic — Complete Hardware Guide

> **Firmware:** Meshtastic · **Upstream:** meshtastic/firmware (GPL-3.0) **Chips:** ESP32-S3 (this profile's default board) · also ESP32, ESP32-C3, ESP32-C6 in the same release family — nRF52840 and RP2040 boards exist but use a UF2 bootloader, not esptool · **Cyber Controller profile:** `meshtastic` (esptool backend, merged single-bin @ `0x0`, no suicide) **This guide:** purchase → build → flash → integrate into Cyber Controller → use → troubleshoot.

1. Overview

Meshtastic is an open-source firmware that turns inexpensive **LoRa** boards into a long-range, low-power **mesh network** for **text messaging, GPS/position sharing, and telemetry — with no cellular, WiFi, or internet required**. Nodes relay each other's packets, so a handful of devices can cover kilometres of terrain; traffic on a channel is end-to-end encrypted with a pre-shared key (AES). It is a *communications* firmware, not an offensive-security tool. Cyber Controller's role is the **flashing** step: it fetches the correct per-board image and writes it over USB. Day-to-day operation (region, channels, messaging) happens through the official Meshtastic **phone app, web client, or Python CLI** — see §6–7.

2. Legal & Safety

Two hard rules before anything else:

1. **Attach the LoRa antenna before powering the board on.** Upstream is explicit: *"Never power on the radio without attaching an antenna as doing so could damage the radio chip!"* Transmitting into an open/unmatched port can destroy the SX126x/LR11xx radio. This applies on first boot and every flash.
2. **Set the correct `lora.region` (it is mandatory and lawful operation depends on it).** Until a region is set, the node displays a warning and **will not transmit any packets**. Each region defines the licence-free ISM band, max power, and **duty-cycle** limits for your jurisdiction — e.g. **US** 902–928 MHz, **EU_868** 869.4–869.65 MHz (duty-cycle limited), **EU_433**, **ANZ** 915–928 MHz, plus CN, JP, KR, TW, IN, RU, and many others (full list in §7 / cite §9).

Lawful framing: Meshtastic runs on licence-free ISM bands, but you must pick **the region you are physically in**, use **a board built for that band** (a 915 MHz board cannot legally operate on 868), and **respect the band/power/duty-cycle rules**. Do not set a foreign region to access a band you are not permitted to use. In most places this is unlicensed operation; some bands (e.g. EU 868) enforce strict duty-cycle — Meshtastic honours these once the region is correct.

3. Hardware & Purchasing

Meshtastic runs on many LoRa boards. The Cyber Controller `meshtastic` profile targets **ESP32-S3 / Heltec-class** boards; pick by how you'll deploy:

Tier	Board	MCU / radio	Why	Where to buy (search terms)
Best starter (this profile default)	**Heltec WiFi LoRa 32 V3** (also V4)	ESP32-S3 + SX1262, 0.96" OLED	~US$20, OLED, battery + (V4) solar/GNSS headers, u.FL LoRa	**heltec.org**, **Rokland**, Amazon/AliExpress: "Heltec LoRa 32 V3 SX1262"
GPS / battery node	**LilyGo T-Beam** (v1.1 / T-Beam Supreme)	ESP32(-S3) + SX1262 + GPS, 18650 holder	Built-in GPS + battery, classic field node	**LilyGo** store, Rokland, AliExpress: "LilyGo T-Beam Meshtastic"
Low-power handheld	**LilyGo T-Echo**	nRF52840 + SX1262 + e-ink + GPS	Very low power, e-ink, GPS — *UF2 flash, not this esptool profile*	LilyGo / Rokland: "LilyGo T-Echo"
Keyboard + screen	**LilyGo T-Deck**	ESP32-S3 + SX1262, QWERTY + LCD	Standalone messaging without a phone	LilyGo / Rokland: "LilyGo T-Deck Meshtastic"
Solar / base station	**RAK WisBlock Starter Kit** (RAK4631 core, RAK19007 base)	nRF52840 + SX1262	Modular, lowest power, best for solar — *UF2 flash, not this profile*	**RAKwireless** store, Rokland: "RAK WisBlock Meshtastic 4631"
Pocket tracker	**Seeed SenseCAP T1000-E**	nRF52840 + LR1110	Card-sized, IP65, GPS — *UF2 flash, not this profile*	**Seeed Studio**, Rokland: "SenseCAP T1000-E"
Compact ESP32-S3	**Station G2 / Nano G2 Ultra** (B&Q)	ESP32-S3 + SX1262	Polished enclosure, optional GPS	B&Q Consulting / Rokland: "Station G2 Meshtastic"

Accessories (essential): a **band-matched LoRa antenna** with the right connector (most boards use **u.FL/IPEX** pigtail to SMA) — buy the **915 MHz** antenna for US/ANZi or **868 MHz** for EU; a **data** USB-C cable (not charge-only); a **18650 / LiPo** for portable nodes; optional **GPS module** and **solar panel** on boards with those headers. **Verify the board's band variant matches your region before buying.**

> ESP32-S3 / ESP32 / C3 / C6 boards flash with this `esptool` profile. **nRF52840 and RP2040 boards (T-Echo, RAK, T1000-E, Mesh Node T114) install via a UF2 drag-and-drop bootloader and are *not* served by the `meshtastic` esptool profile** — flash those with the Meshtastic Web Flasher instead (verify: §9).

4. Building / Assembly

- **Attach the antenna first** (see §2) — then connect battery, then USB. Never key up bare.
- **Pre-built boards (Heltec V3, T-Beam, T-Deck, Station G2):** no assembly — antenna + data cable only.

- **Battery / solar:** plug a LiPo/18650 into the board's battery connector (observe polarity — Heltec uses an SH1.25 JST); V4-class and RAK/Seeed boards expose solar/GNSS headers for off-grid nodes.
- **Drivers (ESP32 only):** install the USB-serial driver for your board's bridge — **CP210x** (most Heltec/ESP32 DevKits) or **CH340/CH9102** (some clones). Native-USB ESP32-S3 boards may enumerate without a driver. Without the right driver, no COM port appears.
- **Per-board gotchas:** confirm the exact variant (e.g. **Heltec V3 vs V2** vs **V4**) — picking the wrong variant flashes the wrong pin map and the OLED/LoRa won't work. nRF52/RP2040 boards have **no serial bootloader** — they mount as a USB drive for UF2 files (not this profile).

5. Flashing & First Run (via Cyber Controller)

1. **Attach the antenna**, then connect the board by USB; open Cyber Controller → **Flash** tab.
2. **Port:** pick the board's COM/tty (click *Refresh* if missing → driver issue, see §4).
3. **Firmware Profile:** `Meshtastic`.
4. **Board / variant:** choose your exact board (default is **`heltec-v3`**). Variants are listed **by chip** — e.g. ESP32-S3: `heltec-v3`, `tbeam-s3-core`, `t-deck`, `t-watch-s3`, `station-g2`, `tlora-t3s3-v1`, `m5stack-cores3`, `seeed-xiao-s3`, and more; ESP32: `tbeam`, `tlora-v2-1-1_6`, `heltec-v2_1`, `rak11200`, `nano-g1`, etc. Pick the one printed on your board.
5. Click **Flash**. Cyber Controller auto-detects the chip, resolves the latest GitHub release, downloads the per-chip `firmware-<chip>-<version>.zip`, extracts the **merged single image `firmware-<board>-<version>.bin`**, and writes it at **offset `0x0`** (8 MB flash, `dio` @ 80 MHz for Heltec V3). If a flash fails, hold **BOOT** while connecting to force download mode (see §8).
6. **First boot:** the OLED shows the Meshtastic splash and a **"region not set"** message — that's expected. The radio stays silent until you set the region (next step, §7).

6. Integrate into Cyber Controller

- **Profile:** `meshtastic` — **backend `esptool`**, `image_model: merged-single-bin`, `app_offset: 0x0`, resolver `github_release` against `meshtastic/firmware` (latest release). Variants are resolved **by chip** from the release zips (`firmware-(esp32|esp32s2|esp32s3|esp32c3|esp32c6)-<ver>.zip`); default variant prefers **`heltec-v3`**. **`supports_suicide: false`** (no self-destruct for this profile).
- **No serial command protocol:** the profile's `protocol` is **`null`** — unlike Marauder, Cyber Controller does **not** parse a Meshtastic serial command set or auto-populate a target pool. The Devices tab can open the raw serial port, but **normal control is the Meshtastic app/web/CLI** (below).
- **Backup first:** use **Backup** in the Flash tab to dump current flash before overwriting; **Erase Flash** is available if you need a clean install (this wipes node config — export it first via the app).
- **Control plane (post-flash):** connect to the node with one of the official clients —
- **Android / iOS app** → pairs over **Bluetooth** (also USB on Android).
- **Web Client** (`client.meshtastic.org`, or the node's `meshtastic.local`) → over **WiFi/USB**.
- **Python CLI** `meshtastic` → over **serial/USB** (or TCP/BLE) for scripting and bulk config.

7. Usage (end-to-end)

1. **Flash** (Cyber Controller, §5) with the antenna attached.

2. **Pair:** open the **Meshtastic phone app** and connect to the node over Bluetooth (or use the Web Client / CLI over USB).
3. **Set region (mandatory):** Settings → **LoRa** → **Region** → choose your country/band (**US**, **EU_868**, **EU_433**, **ANZ**, **CN**, **JP**, **KR**, **TW**, **IN**, **RU**, … or **LORA_24** for 2.4 GHz). The node now starts transmitting. *All nodes that must talk to each other need the same Region and Modem Preset.*
4. **Pick a modem preset:** default **LONG_FAST** (good range/speed balance); options run from **SHORT_TURBO** (fastest, shortest range) to **VERY_LONG_SLOW** (max range, low throughput).
5. **Channels:** the **primary channel** carries your mesh; it's keyed by a **PSK**. Share the channel QR/ URL with others to put them on the same encrypted channel. Add secondary channels as needed.
6. **Message / track:** send text to the channel or a node, view other nodes on the map (GPS-equipped boards report position), and read telemetry (battery, environment) — all from the app/web/CLI.
7. **CLI examples (verify exact flags in §9 docs):** `meshtastic --set lora.region US`, `meshtastic --info`, `meshtastic --sendtext "hello mesh"`.

8. Troubleshooting

- **No COM port:** install the **CP210x/CH340/CH9102** driver; use a **data** cable; on Linux add your user to `dialout` and replug.
- **Flash fails / "Failed to connect":** hold **BOOT** while plugging in (or during connect) to enter download mode, lower the flash baud in Settings, and close any serial monitor/app holding the port.
- **OLED blank or garbled after flash:** wrong **variant** — re-flash selecting the exact board (e.g. V3 vs V2 vs V4).
- **"Region not set" / node won't transmit:** set **LoRa** → **Region** in the app (§7) — this is required by design, not a bug.
- **Nodes don't see each other:** mismatched **Region** or **Modem Preset**, different **channel/PSK**, or **no antenna** on one node. Confirm all three match and antennas are attached.
- **Poor range / radio damaged:** check the antenna is the **right band** and was attached **before** every power-on; an unmatched/missing antenna degrades or destroys the radio.
- **nRF52/RP2040 board won't flash here:** those use a **UF2 bootloader**, not esptool — use the Meshtastic Web Flasher (double-tap reset to mount the drive, drop the UF2). Not served by this profile.
- **Boot-loop / brick: Erase Flash** then re-flash; verify genuine flash size (Heltec V3 = 8 MB).

9. Sources

- Upstream: https://github.com/meshtastic/firmware (GPL-3.0; releases, per-chip firmware zips).
- Docs: https://meshtastic.org — Getting Started & antenna warning (https://meshtastic.org/docs/getting-started/), ESP32 flashing & CLI script (https://meshtastic.org/docs/getting-started/flashing-firmware/esp32/), LoRa region/preset/channels (https://meshtastic.org/docs/configuration/radio/lora/), supported hardware (https://meshtastic.org/docs/hardware/devices/), client apps (https://meshtastic.org/docs/software/).
- Cyber Controller profile: `src/config/profiles/meshtastic.json` (chip/board list, esptool backend, merged-single-bin @ `0x0`, `github_release` resolver, `protocol: null`, `supports_suicide: false`).
- Hardware/purchasing: heltec.org, LilyGo, RAKwireless, Seeed Studio, B&Q Consulting; resellers Rokland, Amazon, AliExpress. **Verify board band variant, current availability, and prices at purchase time; vendor links change.**

- **Verify before relying on exact syntax:** esptool offsets/erase steps for your specific board, current region frequency tables, and `meshtastic` CLI flags — confirm against the upstream docs above.